

Semantic Distinction

Behavioral Consequences and Contextual Neural Differentiation

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Abstract

Franch and colleagues report a positive correspondence between hippocampal population-response distances and contextual language-model embedding distances that changes sign within a restricted range of closely related embeddings. This geometrical observation does not establish an adaptive contrastive mechanism, and neural-model alignment does not establish algorithmic identity. This commentary advances a narrower hypothesis: independently normed, probe-relative consequences of substituting situated interpretations predict neural differentiation beyond embedding, linguistic, and fitted-mapping baselines. A controlled experiment would measure substitution consequences in a separate behavioral sample before neural analysis, then test their association with population distances and held-out discrimination. The proposal distinguishes representational geometry from recoverable information and task consequence, specifying which observations would support contextual differentiation while leaving its biological mechanism open.

Theoretical commentary. No new experimental data or fitted analyses are reported.

1 The problem of preserving relations

Understanding an expression requires recognizing what it shares with other expressions while retaining what makes its present occurrence different. A representation that supports only similarity risks conflating distinctions; a representation that isolates every occurrence loses the regularities through which comprehension becomes possible. Semantic organization consequently involves a tension between generalization and discrimination. This tension becomes especially visible when related expressions occupy nearby positions in an embedding space, yet their substitution changes the interpretation of a sentence. A category can remain stable while the relation asserted within it changes. The difference between an object being available and being unavailable, for example, is small in the vocabulary required to describe it and substantial in the continuation that the description permits.

The hippocampal recordings reported by Franch et al. (2026) provide an empirical occasion for examining this problem. Their study concerns neural responses during narrative listening, and its principal contribution is evidence about population representation rather than a complete theory of meaning. The present paper develops an interpretation of that evidence: semantic geometry may preserve relations through an interaction between contextual integration and selective differentiation. This interpretation is a theoretical extension, not a claim that the experiment directly measured the practical consequences of semantic distinctions.

The explanatory object is contextual discriminability within population representations. Relational memory supplies background motivation for how occurrences might be bound to discourse, but

the proposed experiment does not identify binding by measuring separation alone. “Relational” is therefore used below for the relevant memory literature or for geometrical relations among responses, with those senses distinguished explicitly. The central claim concerns whether a population preserves distinctions required by an independently specified comprehension task. It neither treats relations as an exhaustive account of semantics nor dispenses with reference and the bodily or social conditions of understanding.

1.1 Predictive compression as a theoretical perspective

Tishby, Pereira, and Bialek’s information bottleneck method (1999; archived 2000) formalizes compression of an input X while preserving information about a relevance variable Y . In schematic form, a representation T is selected by minimizing $I(X; T) - \beta I(T; Y)$, with mutual information I and trade-off parameter β . Applied here as a theoretical perspective, X could be the speech history and Y an independently specified future comprehension outcome. Surface differences may then be compressed when they carry no information about Y , while small differences remain relevant when they change its conditional distribution. Neither lexical identity nor acoustic variation is intrinsically irrelevant; relevance depends on Y . This framework motivates the consequence measure without identifying it with mutual information.

The application is conditional rather than a demonstrated optimization process in hippocampus. Compression and predictive retention alone do not determine a Euclidean or cosine geometry, a local correlation sign change, or an evolutionary explanation. Even information-equivalent representations can have different distances under invertible changes of coordinates. A bottleneck account of the reported geometry would additionally require a specified representational family, noise model, relevance variable, and readout. The useful connection to Section 5 is that preserving similarity must be evaluated relative to what a representation needs to predict; it is not a derivation of contrastive separation from efficient coding alone.

For a discrete illustration, mutual information is

$$I(U; V) = \sum_{u,v} p(u, v) \log \frac{p(u, v)}{p(u)p(v)}, \quad \mathcal{J}_{\text{IB}} = I(X; T) - \beta I(T; Y). \quad (1)$$

The sum uses the convention that zero-probability terms contribute zero. Compression concerns dependence between input and representation; predictive retention concerns dependence between representation and the declared relevance variable. If two contexts induce the same conditional distribution of Y , merging their codes need not lose predictive information about that Y . If their conditional distributions differ, merging may incur a predictive loss. This supplies a rationale for task-relative consequentiality, without equating the finite probe score Q with an information-theoretic optimum. In continuous deterministic representations, mutual-information expressions can be infinite or require additional regularity; applying this objective to real-valued neural codes would require an explicit stochastic encoder or another declared regularization. The commentary therefore uses the framework conceptually rather than assigning an unmeasured information rate to the hippocampus.

2 What the hippocampal study establishes

Franch et al. recorded 356 hippocampal neurons in ten native English-speaking patients undergoing monitoring for epilepsy. Participants listened to approximately 47 minutes of narrative speech

containing 7,346 word occurrences and 1,351 unique words. Encoding models related neural activity to linguistic features, including semantic embeddings, with analyses controlling for phonetic and grammatical information. Individual neurons exhibited mixed selectivity across words and semantic categories, while population patterns carried information about contextual semantic organization. These observations support distributed representation under the conditions examined, without requiring every neuron to contribute equally or eliminating more selective responses elsewhere in the brain.

The stimulus consisted of six Moth podcast monologues. Firing-rate epochs began 80 ms after word onset and lasted for that word’s duration, with rates normalized by duration. The Word2Vec Poisson ridge encoding model fit 222 of 356 recorded neurons, or 62%; its median explained variance was 26%, including duration and interaction contributions. The article describes isolated single neurons in its Results, while Extended Data Fig. 1 illustrates single- and multi-unit isolation; a reanalysis should verify the released unit classifications rather than infer their proportions from this illustration. The comparison uses the final BERT representation designated layer 13 and GPT-2 layer 37 in the article’s indexing convention.

The study also compares distances among neural responses with distances among language-model embeddings. Contextual representations from BERT and later GPT-2 layers corresponded more closely to the observed population organization than static Word2Vec representations. Across the full comparison, neural and contextual semantic distances showed small positive correlations. Within a restricted range of especially similar embeddings, the relationship became negative, a pattern the authors interpret as possible contrastive coding. They report that the local reversal was associated with different word pairs rather than repeated occurrences of identical words and that related patterns appeared in analyses conducted separately for individual patients.

The numerical strength and the inferential character of these findings both matter. Overall distance correlations were approximately 0.06 for BERT and 0.09 for GPT-2. The authors estimate that these values capture substantial fractions of the recoverable association under their noise model, but a noise-ceiling estimate is conditional on the assumptions used to construct it. Encoding-model performance also includes contributions from duration and interactions, so total explained variance should not be described as variance attributable exclusively to semantics. Population decoding of semantic embeddings explained approximately 1–6% of variance, depending on the embedding model. These results establish measurable correspondence; they do not establish reconstruction of a listener’s complete meaning representation.

For exact numerical anchoring, Fig. 3d reports BERT $r=0.0635$, and Extended Data Fig. 7a reports GPT-2 layer-37 $r=0.0932$; the Results prose gives nearby values of 0.0650 and 0.0921, respectively. The caption-specific values should not be silently treated as identical to the prose values. The authors report approximately 54% and 76% of their modeled noise ceilings for BERT and GPT-2. Those percentages describe modeled recoverability, not percentages of semantic content decoded or a proof of a shared mechanism. Fig. 3f reports $r=-0.1536$ in the BERT distance interval 0.15–0.35, containing 2.16% of the pairwise observations. This restricted coverage increases sensitivity to range boundaries and composition of the eligible pairs; the pair count does not supply an equivalent number of independent biological observations. The corresponding negative association for different-word pairs is reported as approximately -0.151, while identical-word pairs show a small nonsignificant positive association. Reduced-rank decoding reaches patient-averaged 1-NSE=0.06 for GPT-2, with a median optimal rank of 12; neither number should be read as a complete dimensionality estimate of hippocampal semantics.

Frequency and polysemy also require distinctions between analyses. High-frequency categories are centrally located in the two-dimensional projection while showing greater separation in the

full population space. In the regression predicting distance across occurrences of the same word, polysemy and duration are significant predictors. The separate pair-distance regression includes semantic distance, frequency, duration, category, and polysemy differences. These results concern particular distance outcomes, not an unrestricted claim that polysemy and duration explain all neural variance.

The authors explicitly acknowledge that controlled follow-up studies are needed to test their contrastive interpretation. That acknowledgment sets the appropriate evidential boundary for this commentary. The distance reversal is an observation; a mechanism that increases separation to reduce semantic confusion is an explanatory hypothesis. The two deserve different verbs and different standards of confirmation.

The local negative association is a departure from the modest positive associations just reported, rather than a reversal of an otherwise near-perfect correspondence. Its interpretation depends on the definition of the local range. The article reports approximately 0.15–0.35 for a focal BERT analysis and approximately 0.14–0.4 for related analyses; these numerical cosine-distance ranges refer to the particular comparison embeddings and do not define a universal threshold of semantic similarity. A selected range can emphasize curvature or a local fluctuation, especially when its location and endpoints are chosen after viewing the association. The commentary therefore treats the reported sign change as a candidate scale-dependent pattern whose stability should be tested across independently specified ranges, rather than as an established boundary between two coding mechanisms.

3 Relational representation in the existing literature

3.1 Relational memory and language use

Konkel and Cohen (2009) characterize hippocampal memory through binding and flexible retrieval of spatial, sequential, and associative relations; Duff and Brown-Schmidt (2012) extend relational binding and representational flexibility to discourse and shared knowledge. These accounts motivate investigating contextual integration during language use without treating distances between word-evoked responses as measurements of relational binding. Geometry among responses and memory for experienced relations remain connected hypotheses rather than interchangeable observations. The present experiment tests contextual discriminability; identifying a binding contribution would require a separate relational manipulation.

3.2 Separation and integration as complementary demands

McClelland, McNaughton, and O'Reilly (1995) formulate complementary learning systems as an account of how rapid acquisition of specific experiences can coexist with the gradual extraction of shared structure. Their computational argument identifies interference as a problem for learning systems and motivates different representational and learning properties in hippocampus and neocortex. This provides a precedent for the tension developed here, although the present question concerns differentiation during contextual language processing rather than the complete architecture of consolidation. The commentary does not introduce the general problem of preserving specificity while learning regularities.

Bakker et al. (2008) use high-resolution functional imaging to examine responses consistent with pattern separation in the combined human CA3/dentate-gyrus region. Their evidence links

similar inputs to hippocampal processing that distinguishes them, but its measurement scale and stimulus paradigm differ from single-neuron responses during narrative listening. Pattern separation is conventionally understood as a transformation of input similarity into less overlapping output representations. A negative association between neural distances and distances in a comparison embedding is not by itself a direct measurement of that transformation, since the embedding is not a recording of the hippocampus’s biological input. Anatomical localization also matters: a hippocampal population result does not alone establish a dentate-gyrus mechanism.

Schlichting, Mumford, and Preston (2015) provide a particularly close precedent for treating integration and differentiation jointly. Their study of overlapping associations identifies learning-related integration and separation signatures in different hippocampal and prefrontal regions, with learning conditions influencing the outcome. The result cautions against assigning a single representational policy to the entire hippocampus. It also motivates a distinction between the separation of events that share elements and the integration of elements that connect events. The proposed semantic account extends this question to occurrences in discourse; it does not claim that a word-pair distance reversal identifies the same process or reproduces the same regional dissociation.

Together, these precedents constrain the interpretation of “contrastive.” It can name a descriptive geometrical pattern, a hypothesized input–output transformation, or an adaptive learning process that changes representations over time. These meanings require different evidence. The Franch et al. result supplies a candidate descriptive pattern; the experiment proposed below would test its sensitivity to contextual demands. Establishing a learning mechanism would additionally require observations of representational change under controlled exposure or intervention.

3.3 Mixed selectivity and population readout

Rigotti et al. (2013) show that nonlinear mixed selectivity in monkey prefrontal cortex can support high-dimensional representations and flexible readout during a sequence-memory task. Their analyses connect population dimensionality to behavioral performance. This establishes a computational precedent for information residing in combinations of responses that are not easily interpreted neuron by neuron. It does not make mixed selectivity a sufficient condition for semantic representation, and selectivity across several word categories should not automatically be equated with the particular nonlinear interactions analyzed in their task. The hippocampal findings motivate a population account, while additional analyses would be needed to establish which interaction structure supports contextual discrimination.

Mixed selectivity thus motivates a population-level inquiry, while its functional interpretation requires the noise-aware readout and independent behavioral criterion specified in Section 5.2.

3.4 Contextual meaning within a wider semantic architecture

Deniz et al. (2023) show that linguistic context affects semantic representations and their measurement across words, sentences, and narratives, motivating the situated-occurrence analysis without equating every context effect with polysemy. Patterson, Nestor, and Rogers (2007) locate conceptual knowledge within a distributed architecture involving modality-specific representations and a transmodal anterior temporal hub. These precedents constrain the hippocampal interpretation: contextual differentiation can coexist with relatively enduring conceptual structure, and hippocampal coding does not make that region the sole repository of meaning. Neither architecture nor contextual variability alone predicts the local distance sign change; the operational question is which independently consequential distinctions remain recoverable.

3.5 Language-model correspondence and its evidential level

Goldstein et al. (2022) report correspondences between neural responses during narrative comprehension and autoregressive language models, including contextual representation, anticipatory prediction, and responses associated with surprise. Their work establishes precedent for using naturalistic language-model predictions to investigate neural processing. These are specified computational correspondences under the analyses performed, not evidence that every operation or learning process is shared. The hippocampal result should be interpreted within that comparative program while preserving its distinct anatomical target and measurement scale.

Goldstein et al. (2024) examine contextual embedding geometry using dense intracranial recordings from the inferior frontal gyrus of three participants listening to a podcast. Their zero-shot mapping analyses relate neural representations of held-out words to language-model geometry and find an advantage for contextual over static embeddings. This provides a direct precedent for studying common geometrical patterns in language representations. Franch et al. extend the comparison to hippocampal single-neuron populations and identify a scale-dependent discrepancy. The difference is theoretically informative, but the regional and recording differences prevent treating it as a controlled comparison of hippocampal and frontal coding policies.

The explicit puzzle is that contextual embeddings outperform static representations in both studies, whereas the hippocampal paper reports a local sign change that is not established as a matched frontal result in this comparison. Is this discrepancy related to hippocampal differentiation, or would equivalent sampling and local-distance analyses reveal it in frontal responses as well? Lack of a matched report is not evidence of absence. A regional claim requires the same stimuli, comparable measurement and preprocessing, and prespecified range analyses in both regions.

The contribution of the present commentary is correspondingly narrower than the discovery of vectorial or contextual semantics. It proposes that independently measured consequences of confusion could help explain when geometrical separation becomes useful. Neither shared geometry nor disagreement establishes that explanatory variable; it must be manipulated or measured outside the embedding and neural-distance analyses.

3.6 Representational similarity and the limits of geometrical inference

Kriegeskorte, Mur, and Bandettini (2008) introduce representational similarity analysis as a framework for comparing relations among activity patterns with computational and behavioral models. Its relevance here is that correspondence among distance structures concerns relations within representations, even when individual measurement channels have no direct counterparts. They also emphasize that distances sharing observations are dependent and motivate inference through condition-label randomization. This precedent supports treating geometry as an object of inquiry while requiring uncertainty estimates that respect the structure of the observations. It also makes independent behavioral comparison central to the proposed functional interpretation.

A related methodological point follows from considering a linear mapping $r = Az$ from contextual embeddings z to neural responses r . Even without an additional term, a general A need not preserve cosine distance or local distance order. A geometrical discrepancy can therefore occur without an additional separation mechanism. Evidence for selective differentiation requires comparison with mappings that already allow such distortions, followed by evaluation on held-out conditions. Otherwise, an unexplained residual risks naming whatever the initial mapping fails to capture. The strongest version of the proposal asks whether contextual confusion consequences explain reproducible structure beyond suitable contextual, linguistic, and mapping baselines.

This focused narrative review identifies precedents and evidential boundaries rather than claiming exhaustive coverage. Its synthesis is that relational binding, mixed selectivity, context dependence, and the integration–separation tension are established research themes. The open question advanced here concerns their conjunction in semantic discrimination: whether the practical consequence of confusing situated occurrences predicts population differentiation beyond the comparison embedding distance.

3.7 Coordinates, metric structure, topology, and readout

Recent work gives empirical motivation for separating coordinate-level tuning from population organization. Yan et al. (2026) report shared semantic geometry across English and Spanish in human hippocampal recordings despite differences in individual-neuron tuning, with language-specific readout axes within a common recorded population. Their result supports distinguishing metric correspondence from coordinate identity. The readout axes identified analytically do not themselves establish which downstream biological circuit decodes those responses, and the authors’ interference interpretation remains a functional hypothesis. Shared cross-language geometry also does not demonstrate the consequence-dependent differentiation proposed here.

Topology supplies a complementary description rather than another name for distance. Brown and Farivar (2025) use persistent homology to characterize components, loops, and higher-dimensional structure across distance thresholds, showing why high correlation between dissimilarity matrices can conceal differences in shape. This does not mean that a complete distance matrix contains no topological information: their analysis itself uses distance relations. The limitation concerns the scalar comparison applied to those matrices. Coordinates, the declared metric, topology of the sampled or hypothesized support, and recoverability under a readout should therefore be distinguished as structures with different invariances, rather than treated as a strict ladder of explanations. None alone supplies an independent measure of comprehension consequence.

4 Situated occurrences as representational units

A lexical type provides a convenient index for grouping occurrences, but grouping is not an explanation of how those occurrences are understood. The same expression can acquire different relations through its position in a narrative, its grammatical environment, and the expectations established by preceding speech. Franch et al. examine this contextual variability through both embedding-based measures and a WordNet measure of polysemy. Their findings associate variation across occurrences with variation in neural responses. The distinction between an embedding-based contextual measure and a lexical inventory of senses remains important, since contextual variability can occur without a clean division into dictionary senses.

The theoretical consequence is a change in the unit of analysis. Instead of beginning with a representation $r(w)$ assigned once to word w , consider a representation $r(w,c)$, where c denotes the context relevant to that occurrence. This notation does not explain how context is computed, how much preceding material is retained, or whether different neural populations contribute distinct contextual variables. It merely prevents lexical identity from being mistaken for representational identity. Two occurrences of the same word may remain recognizably related while differing in their encoded contribution to the current discourse.

More precisely, write the occurrence as $o_i=(w_i,t_i,e_i)$, identifying its lexical type, onset time, and narrative episode, and write $c_i=C(H_{t_i},e_i)$, where H denotes the linguistic history available

before or during the measured response. Context is not assumed to be the entire recording session, nor is it restricted in advance to one sentence. Candidate operationalizations should distinguish preceding words, sentence-level information, and the currently maintained narrative episode, with separate variables for prior knowledge and experimentally supplied task instructions. The present commentary brackets the biological computation of C and the duration of its effective memory; it proposes comparing these candidate scopes. Model context windows and biological integration timescales must also remain distinct, since a model window is an analytical choice rather than evidence that a listener retains exactly that interval. Within each tested condition, c_i should be specified from the stimulus and design independently of the neural outcome.

Context sensitivity also changes the interpretation of averaging. An average over occurrences can reveal a stable lexical component, yet it can obscure systematic contextual distinctions. Conversely, variation among occurrences need not all be meaningful: recording noise, adaptation, timing, and unmeasured narrative variables can contribute. A relational analysis should therefore seek reproducible context-dependent structure within the variation, rather than treating either the average or the variability as automatically authoritative.

4.1 Mathematical objects and distinction preservation

The operational hypothesis requires interpretation, comparison representation, response variability, and task outcome to be specified separately, so the following vocabulary fixes their domains before any distance or consequence is compared. Let $\mathcal{O}$ be the set of situated occurrence records, $\mathcal{K}$ the experimental conditions, $\mathcal{M}$ a declared interpretation space, $\mathcal{Z}$ the comparison-representation space, and $\mathcal{Y}$ the task-outcome space. An occurrence record $o_i = (w_i, t_i, e_i)$ identifies a token through its lexical type, onset, and episode; its analytical context $c_i^{(k)}$ is supplied by the history and design under condition k . Distinct occurrence records may share a lexical type. For the proposed experiment, $\mathcal{M}$ is a finite set of explicitly specified discourse interpretations admissible for the stimulus, such as alternative assignments of a referent or relation. Each interpretation is an experimental description with declared propositions and probe-answer implications, rather than an identified biological state or a complete theory of meaning. $P(M \mid o_i, k)$ records uncertainty over those alternatives, including an indeterminate category when the protocol permits it. Equality of interpretation labels, selection of an admissible alternative, and a protocol-defined substitution are permitted operations; no intrinsic metric on $\mathcal{M}$ is assumed. The operational measurements of Y and Q are more developed than this interpretation space, intentionally, because the hypothesis does not require exhaustive semantic states. Neither an interpretation label nor lexical identity determines a unique neural response. Lexical non-identification is the relevant principle: shared lexical type does not determine shared context, interpretation distribution, or population response.

The comparison encoder and neural measurement have different domains and noise commitments:

$$E : \mathcal{O} \times \mathcal{K} \longrightarrow \mathcal{Z}, \quad \mathcal{R}_s : \mathcal{O} \times \mathcal{K} \longrightarrow \mathcal{P}(\mathbb{R}^{n_s}), \quad r_{si}^{(k)} \sim \mathcal{R}_s(o_i, k). \quad (2)$$

Here $\mathcal{P}(\mathbb{R}^{n_s})$ denotes response distributions for participant s , while $z_i^{(k)} = E(o_i, k)$ is the model representation under declared preprocessing. Neural distances compare realized or prespecified estimated response vectors within a participant, whereas readout discrimination depends on their conditional distributions. These are parallel measurements of an occurrence. There is no assumed causal chain from interpretation to embedding to recorded neurons, and no assumption that neural coordinates correspond across participants.

A probe protocol π specifies the questions and their weighting, answer criterion, norming

population, and aggregation rule. Write $\rho_i^{(k,\pi)} = P(Y \mid o_i, k, \pi)$ for its independently characterized outcome distribution. A declared divergence between two such distributions defines a task-predictive distinction, not a geometric distance between representations. For example, total variation gives

$$\Delta_{Y,\pi}(i, j; k) = \frac{1}{2} \sum_{y \in \mathcal{Y}} \left| \rho_i^{(k,\pi)}(y) - \rho_j^{(k,\pi)}(y) \right|. \quad (3)$$

For a neural readout g_s that returns an outcome distribution, define its averaged prediction $\hat{\rho}_i = \mathbb{E}[g_s(R) \mid o_i, k]$. Relative to the declared pair and task, an approximate preservation criterion is $\text{TV}(\hat{\rho}_i, \rho_i) \leq \varepsilon$ at each endpoint. The triangle inequality then gives $\text{TV}(\hat{\rho}_i, \hat{\rho}_j) \geq \Delta_{Y,\pi}(i, j; k) - 2\varepsilon$. This is a sufficient approximate criterion for preserving the declared outcome distributions, not a necessary-and-sufficient characterization of all semantic discrimination. When the task-predictive distinction exceeds twice the error tolerance, the predicted distributions must remain distinct, without prescribing a Euclidean distance. It must be evaluated using a declared readout class and held-out observations; accurate population-average prediction alone need not preserve every rare distinction. The expectation averages response variability: matching outcome distributions on average does not establish successful trial-wise interpretation classification. Section 5.2 gives an illustrative noise-aware classification analysis, and Section 8 specifies separate held-out tests of distribution preservation and discrimination.

The substitution score in Section 8 is a separate functional of a protocol and directional intervention. Write it fully as $Q_{ij}^{(k;\pi)}$, with $Q_{ij}^{(k)}$ used as shorthand only when π is fixed. A substitution operator must specify which interpretation is introduced into which receiving context, and the independently normed comparison must be assigned to occurrence pairs before neural analysis. The score records probe-relative answer changes under that intervention; it is not automatically the total variation between outcomes of two unmodified occurrences. Nor is it confusion susceptibility, severity, or a universal semantic distance. This distinction prevents the behavioral score from being inferred from neural geometry or imported from an item pair without a declared occurrence assignment.

Functional equivalence is consequently relative to a task and readout class. Two representation–readout pairs are equivalent for the declared task when they generate the same conditional outcome predictions over the eligible domain. Invertible reparameterizations preserve that equivalence when the transformed readout remains in the allowed class, but generally change coordinate distances. This observation unifies the linear-map and CPC examples without asserting equivalence under every biologically restricted readout. Geometry, predictive information, recoverability, behavioral consequence, and randomized assignment effects are distinct evidential objects; the empirical hypothesis links them rather than identifying them.

4.2 Multiple temporal horizons

Lerner et al. (2011) use scrambled narrative speech to investigate a hierarchy of cortical temporal receptive windows. Their results motivate distinguishing sensitivity to local speech structure from integration over longer narrative intervals. Baldassano et al. (2017) investigate event structure in continuous narratives, linking cortical event representations with hippocampal responses around event boundaries. These precedents support testing multiscale and event-dependent context, without establishing a particular decay law for hippocampal semantic responses.

A candidate analytical representation is $c_i = (h_1(t_i), \dots, h_L(t_i), s(e_i))$, where $h_l(t)$ integrates features of preceding input under a kernel K_l and s represents episode-level information. An exponential

kernel is a parsimonious baseline because it represents recency weighting with one timescale and permits recursive updating with a finite state; neither property establishes its biological adequacy. Power-law mixtures and boundary-gated kernels supply alternatives to compare on held-out material. For example, an exponential kernel with timescale τ_l supplies a testable baseline; an event-gated kernel can reduce carryover at an independently annotated episode boundary. These are candidate predictors, not identified neural mechanisms. Region-specific combinations of horizons could be investigated along the hippocampal long axis or in hippocampal–cortical networks, including regions associated with the default mode network. The model must not assume that anterior and posterior populations instantiate particular timescales solely from their anatomy. Comparisons should fit kernels on training material and test them on held-out episodes, keeping independently annotated boundaries separate from boundaries estimated using the neural outcome.

For a feature stream $\phi(t)$, one such normalized history predictor is

$$h_l(t) = \int_0^\infty K_l(u)\phi(t-u) du, \quad K_l(u) = \tau_l^{-1}e^{-u/\tau_l}, \quad \tau_l > 0. \quad (4)$$

For a finite recording, the stream must be defined before its beginning or the truncated kernel renormalized. An episode gate $g(t,u)$ can set contributions preceding an independently annotated boundary to zero, replacing the kernel by $K_l(u)g(t,u)$ divided by its integral. Exponential decay encodes a particular recency assumption; abrupt gating encodes a different assumption about boundary-related updating. Comparing their held-out performance would test descriptive temporal models, not identify the biological storage process. Identifiability also depends on stimulus variation: highly correlated horizons may not support separate timescale estimates, even if their combined predictor improves fit.

4.3 Event scripts, task framing, and hierarchical recall

Research on event scripts motivates enriching context beyond elapsed time. Baldassano, Hasson, and Norman (2018) identify schematic sequences shared across restaurant and airport narratives. De Soares et al. (2024) examine overlapping location and social scripts and show that top-down attention shifts behavioral segmentation and the alignment of medial prefrontal event boundaries. These findings motivate distinguishing stimulus history from the script made relevant by instructions. They do not establish that hippocampal word-distance changes arise through the same mechanism.

An extended analytical context is $c_i = C(H_{t_i}, e_i, S_i, G_i)$, with S denoting a candidate script and G task framing. Script identity, temporal horizon, and segmentation are distinct variables: listeners can hear the same history while weighting different transitions as relevant. A controlled experiment could hold the recording constant while varying location-focused or socially focused instructions. Consequence norming must use the corresponding frame, since relevance depends on the questions asked. Frame effects may also reflect attention and retrieval demands, which must remain alternative explanations.

Zhong et al. (2025) provide a complementary account of narrative compression. Its random-tree model represents narratives through hierarchies of summarized segments and constrains retrieval by working-memory capacity. It addresses quantitative regularities in recall length and the narrative portions summarized by individual recall clauses. A model tree node is not an identified hippocampal state or an information-bottleneck optimum. The account helps distinguish abstraction during recall from temporal integration during listening and preservation of task-relevant distinctions. Agreement with aggregate recall statistics does not determine word-level neural geometry.

5 Similarity and discriminability

An embedding distance can be useful without functioning as a universal ordering of substitutability. Two expressions may be close because they concern the same participants, objects, or circumstances while differing in a relation that comprehension must preserve. This observation motivates a distinction between broad semantic affinity and contextual discriminability. The former supports transfer and association; the latter determines whether a listener can retain a difference required by the present interpretation.

The reported local reversal is compatible with this distinction. If closely related meanings would otherwise produce responses that are difficult to distinguish under biological noise, additional differentiation could preserve their identity without abolishing their broader affinity. The argument is functional rather than demonstrative: it identifies a problem that contrastive coding could solve. The observational result does not prove that neural responses were moved apart by an adaptive process, that the relevant words were behaviorally confused, or that separation improved comprehension.

Several explanations can coexist at this stage. Contextual binding could create differences that a particular embedding compresses. The measured distance could change with normalization or representational dimensionality. Different linguistic properties could contribute residual structure despite the reported controls. A mechanism supporting memory differentiation could also produce a semantic pattern without having been specialized for lexical discrimination. Treating the result as a question about these alternatives makes the theoretical claim more useful: it directs inquiry toward the conditions under which local separation occurs and what that separation accomplishes.

5.1 Distance distortion and an explicit counterexample

The distinction between geometrical correspondence and mechanism can be made exact. For $z \in \mathbb{R}^d$ and $r \in \mathbb{R}^n$, let $A \in \mathbb{R}^{n \times d}$ define the linear map $r = Az$. Squared Euclidean response distance becomes

$$\|r_i - r_j\|_2^2 = (z_i - z_j)^\top A^\top A (z_i - z_j). \quad (5)$$

Thus a linear mapping induces a positive-semidefinite quadratic form $G = A^\top A$ on embedding differences. If $G = aI$ for a positive scalar a on the relevant subspace, Euclidean distances are uniformly rescaled and their ordering is preserved. A general G stretches directions unequally; distances of two pairs can change order because their difference vectors point in different directions. This is a property of the mapping rather than evidence of an added differentiation process.

For a concrete example, take embedding difference vectors $\delta_1 = (0.1, 0)$ and $\delta_2 = (0, 0.2)$, and let $A = \text{diag}(3, 1)$. Their original distances are 0.1 and 0.2, but mapped distances are 0.3 and 0.2. The more similar pair becomes more distant in the response space under a fixed linear transform. This example establishes the possibility of an ordering change, not a reproduction of the article's distribution-wide local correlation. Replicating that correlation would additionally require a specified distribution of pairs and measurement noise. It nevertheless demonstrates why an observed local ordering discrepancy cannot identify an adaptive separation mechanism on its own.

For nonzero vectors, cosine distance after mapping is

$$d_{\cos}(Az_i, Az_j) = 1 - \frac{z_i^\top G z_j}{\sqrt{z_i^\top G z_i} \sqrt{z_j^\top G z_j}}. \quad (6)$$

The same induced quadratic form changes angles as well as lengths. On the represented subspace, a positive scalar multiple of an orthogonal transformation preserves cosine geometry; arbitrary linear maps do not. Additive baselines and nonlinear transformations can produce further changes. A consequence-augmented model must therefore be assessed against mapping baselines of declared complexity, with regularization and out-of-sample evaluation preventing a flexible mapping from merely memorizing the tested pairs.

An invertible linear transformation is a homeomorphism of the ambient Euclidean space, so it preserves the ordinary topology of a represented support while potentially changing its metric geometry. Persistent-homology birth and death scales can nevertheless change because the filtration depends on the selected distances. Conversely, projection or undersampling can obscure a support’s topology. Brown and Farivar’s (2025) analysis consequently motivates an additional testable description, not a topological explanation of the reported local sign change. A topology claim here would require a specified support, sampling assumptions, filtration, and uncertainty analysis; no such claim follows from the distance correlation alone.

5.2 Separation, noise, and usable discrimination

Distance has a functional interpretation only under assumptions about response variability and readout. Consider an illustrative binary discrimination problem in which response R conditioned on interpretation a or b follows a multivariate Gaussian with means μ_a and μ_b and a common positive-definite covariance Σ . With equal class priors, the optimal linear discriminant depends on the squared Mahalanobis separation

$$\delta^2 = (\mu_a - \mu_b)^\top \Sigma^{-1} (\mu_a - \mu_b). \quad (7)$$

Under these assumptions, its minimum classification error is $\Phi(-\delta/2)$, where Φ is the standard normal cumulative distribution function. To see this, project R onto $\Sigma^{-1}(\mu_a - \mu_b)$. The projected class means differ by δ^2 and their common variance is δ^2 ; a midpoint threshold therefore lies $\delta/2$ standard deviations from either mean. Increasing δ reduces error in this model. Increasing raw Euclidean distance need not do so if the separation grows along a direction of high or increasing noise. Likewise, two populations with the same cosine distance can differ in discriminability because their covariance structures differ.

This calculation is an analytical example, not a Gaussian model fitted to hippocampal spiking. Neural spike counts may have mean-dependent variance, correlated fluctuations, and temporal dependence. The example’s purpose is to identify the missing bridge between geometry and function: a readout-relevant signal-to-noise quantity and an independent behavioral criterion. A future experiment should evaluate discrimination using held-out responses or suitably regularized covariance estimation. Covariance parameters must be estimated without using test observations, especially when recorded-unit dimension is large relative to repeat count. Single presentations provide limited information about condition-specific response distributions; controlled repetitions would strengthen this part of the functional test.

5.3 Predictive distinction without a prescribed geometry

The information bottleneck gives a normative interpretation of relevant distinction: its distortion compares conditional outcome distributions rather than points in an embedding or neural space. Under the declared encoder assumptions, its expected KL distortion is lost predictive information,

or excess expected log loss. The stationary-encoder condition favors codes whose predictions approximate those required by the input, but does not require a particular Euclidean or cosine separation. Appendix A derives this relation and illustrates the compression trade-off with a binary example. Predictive distortion, embedding distance, and neural representational distance remain different mathematical quantities. Likewise, IB code bifurcations do not identify the reduced-rank predictive dimensions reported by Franch et al.; relating them would require an additional fitted model.

6 Operational hypothesis

The central hypothesis is that increasing independently normed, probe-relative substitution consequence predicts greater neural representational differentiation beyond lexical, contextual-embedding, task-difficulty, and fitted-mapping baselines under a controlled manipulation. Susceptibility to making that substitution is a separate behavioral quantity, rather than part of the definition of Q . Section 8 specifies how both are measured before neural analysis. An adequately powered failure to find the predicted differentiation would count against this functional extension even if the original local sign change replicated. This is an operational hypothesis, not an optimization model or a derivation of the reported sign change; no loss function is needed to state its independent measurement requirement.

For participant s and comparison condition k , let $r_{si}^{(k)}$ be the recorded population response to occurrence i , and let $z_i^{(k)}$ be its contextual embedding. Define

$$D_{r,sij}^{(k)} = d_r(r_{si}^{(k)}, r_{sj}^{(k)}), \quad D_{z,ij}^{(k)} = d_z(z_i^{(k)}, z_j^{(k)}), \quad i < j. \quad (8)$$

Each neural distance is computed within a participant’s population; neurons from different participants are not assumed to be corresponding coordinates. Neural and embedding metrics need not be identical, but their preprocessing, handling of zero vectors, and scaling must be prespecified. Cosine distance is one candidate for nonzero vectors. Euclidean distance requires a declared scale and normalization policy. Let $Q_{ij}^{(k)}$ denote the symmetric consequence score assigned from independent norming to an eligible item pair and condition. Averaging the two directional substitution scores must precede analysis of a symmetric neural distance; directional scores could instead support a separate directional behavioral analysis. Only pairs with a defined normed comparison enter the consequence analysis. Q is not imputed to every pair in a narrative merely because those occurrences have embeddings.

Thus $r_{si}^{(k)}$ belongs to $\mathbb{R}^{n_s}$, where n_s is that participant’s recorded unit count; the total of 356 across the published sample is not the dimension of every participant’s response vector. Cosine distance and the article’s word-aligned response window provide replication-oriented defaults, with alternative metrics and windows declared as sensitivity analyses. The published 0.15–0.35 BERT range can define a replication target under matching preprocessing, but its transfer to a new stimulus set or embedding model requires independent calibration.

Before choosing a regression, define the associational target. Let W collect the declared embedding, linguistic, task, and mapping baselines for an eligible comparison, with a prespecified participant–stimulus weighting distribution. Where a differentiable conditional mean is meaningful, write

$$\mu(q, w) = \mathbb{E}[D_r \mid Q = q, W = w], \quad \tau_Q(q, w) = \frac{\partial \mu(q, w)}{\partial q}. \quad (9)$$

The hypothesis predicts a positive conditional consequence association in the declared regime. This derivative is not a causal effect of setting the behavioral score. Because the finite probe score has discrete support, a prespecified difference in conditional means between supported score levels is an alternative estimand; a smooth derivative then reflects a working model, not an assumption that every intermediate score is experimentally realizable. Baselines, support, and target weighting must be specified before the estimator and test are chosen.

A candidate regression for those eligible pairs is

$$\begin{aligned} D_{r, sij}^{(k)} = & \alpha + f(D_{z, ij}^{(k)}) + \theta M_{sij}^{(k)} + \gamma^T \mathbf{X}_{sij}^{(k)} \\ & + (\beta_Q + b_{Qs}) Q_{ij}^{(k)} + \beta_R I_{ij}^{(k)} + \beta_{QR} Q_{ij}^{(k)} I_{ij}^{(k)} \\ & + b_{0s} + a_i + a_j + v_{si} + v_{sj} + \epsilon_{sij}^{(k)}, \end{aligned} \quad (10)$$

where $I_{ij}^{(k)}$ indicates a prespecified local embedding-distance range and f is a prespecified smooth baseline, such as a penalized thin-plate regression spline. M is the distance predicted by a contextual-embedding-to-neural mapping fitted on independent training data. The nuisance vector $\mathbf{X}$ includes duration contrasts and, where appropriate, pair means, phonetic distance, declared syntactic features, frequency, temporal separation, and independently normed susceptibility and difficulty. A syntactic tree distance must be operationally defined for occurrences in different sentences rather than presumed to have a unique value. Smoothness, feature selection, mapping complexity, and scaling are fixed or selected within training folds. Strong collinearity among these terms limits the estimability of a unique consequence association and must be reported.

The participant intercept b_{0s} and consequence slope b_{Qs} may be modeled jointly as mean-zero Gaussian effects with an estimated covariance. Shared stimulus-occurrence effects a_i capture item variation across participants, while participant-specific occurrence effects v_{si} capture repeated use of a recorded endpoint in that participant’s distances. The same occurrence effect contributes at either endpoint, preserving symmetry under exchange of i and j . These effects are candidate covariance components rather than a guarantee of independent residuals: distances also share population measurements and temporal structure. The estimable random-effects structure should be preregistered subject to declared simplification rules, with design-based inference and simulation-based calibration used to assess remaining dependence.

Without the interaction, the proposed conditional association is $\beta_Q > 0$. With the interaction, the outside-range slope is β_Q and the inside-range slope is $\beta_Q + \beta_{QR}$. Enhanced local differentiation requires $\beta_{QR} > 0$; positive differentiation within the local range separately requires $\beta_Q + \beta_{QR} > 0$. A positive interaction alone is insufficient if the total local slope remains negative. Conversely, a positive local consequence slope does not demonstrate a negative embedding-distance slope. The latter concerns the derivative of the embedding-distance relationship, with other predictors controlled, and must be tested separately. A continuous alternative replaces $\beta_R I$ with $h(D_z)$ and $\beta_Q Q + \beta_{QR} QI$ with $Qg(D_z)$, using prespecified smooth functions and simultaneous uncertainty bands. It avoids hard boundaries but still requires a declared criterion for stronger consequence effects at high similarity.

Evidence for added predictive information would additionally require improvement over the same baseline without Q and its interaction on held-out stimulus blocks. All pairs involving a held-out occurrence must be excluded from training when testing generalization to new occurrences.

A random split of pairwise rows would leak shared endpoints across folds. These regressions specify estimands and comparisons; no coefficients are estimated in this commentary, and an adjusted consequence association alone would not establish causality.

6.1 Identifiability and held-out prediction

The conditional estimand requires overlap: eligible comparisons with different consequence scores must exist at sufficiently similar baseline features. If the consequence score is determined by those features, a partial association cannot be identified by adding more controls. The regression is one working model for the estimand, rather than its definition. Nuisance adjustment must address both the outcome and the consequence predictor, while uncertainty must accommodate shared endpoints and participants. Appendix B gives the elementary projection identity and a held-out predictive-gain statistic. The substantive requirements are a declared direction of association, predictive improvement on genuinely unseen occurrences, and an independent discrimination outcome; success on one does not substitute for the others.

6.2 Conditional association and randomized intervention

Randomization identifies the effect of the assigned context manipulation under its design assumptions, not automatically the causal effect of its derived consequence score. Let K be a randomized context assignment and $D_p(k)$ the potential neural distance for an eligible comparison under assignment k . Its design-based target is $\tau_K = \mathbb{E}[D_p(1) - D_p(0)]$ under the prespecified comparison weighting. This assignment effect and $\tau_Q(q, w)$ are different estimands and have no general equality. Randomization supports estimation of this contrast when assignments, interference assumptions, and missingness are handled appropriately. It does not establish that every effect of K passes through Q , because context can also change attention, prediction, acoustic interpretation, or other semantic variables.

Accordingly, the experiment should separate a primary assignment contrast from the adjusted Q association. Conditioning on variables caused by K , such as a context-sensitive surprisal measure, changes the estimand and may remove part of the total assignment effect. A mediation interpretation would require further identification assumptions and a dedicated design. The commentary’s principal prediction remains that consequential distinctions are associated with additional usable differentiation under controlled conditions, not that a regression coefficient uniquely measures an adaptive biological mechanism.

7 Model alignment and predictive-learning comparisons

Language-model embeddings provide a tractable comparison space, but their usefulness should not be confused with their authority as a definition of meaning. Neural alignment shows that aspects of one measured organization can be related to aspects of another. It does not establish shared training objectives, equivalent learning histories, identical memory operations, or equivalent relationships to action and experience. Franch et al. explicitly distinguish shared geometry from shared algorithm.

The comparison is particularly informative when it reveals disagreement. A model that captures broad relations while missing a local pattern can help identify which distinctions its geometry underrepresents relative to the neural measurement. Yet disagreement alone does not determine which representation is functionally superior. Superiority requires an independently specified criterion, such as discrimination performance or successful contextual interpretation. Likewise,

stronger alignment by one model among those tested establishes a comparative result within that analysis, rather than a general ranking of architectures as theories of cognition.

The broader case for vector representations developed by Piantadosi et al. (2024) makes such comparisons theoretically intelligible: vectors can support similarity, relations, and compositional operations within a common representational framework. The hippocampal findings sharpen a further question about how these capacities are reconciled when generalization and differentiation impose competing requirements. Representational format leaves open the processes that construct and use it.

Fodor and Suzuki (2026) review 57 fMRI studies using computational semantic models and identify substantial methodological variation, alongside robust model–brain associations often dominated by word-level features. Their synthesis supports asking what measured feature explains correspondence before interpreting it as a semantic mechanism. Because their review concerns fMRI rather than hippocampal single-unit geometry, it supplies a methodological precedent rather than direct evidence for the present consequence hypothesis.

Cheng, Vaidya, and Antonello (2026) connect intermediate-layer brain predictivity in language and speech models with intrinsic dimensionality and semantic abstraction, using fMRI and ECoG comparisons as well as model interventions. Their results motivate testing representation properties alongside predictive performance. They do not establish that prediction is irrelevant to biological comprehension or identify the origin of the hippocampal local sign change. The distinction is between an objective that trains a model and the properties of its learned representation that explain a particular alignment result.

7.1 Contrastive learning as a structural comparison

Contrastive predictive coding (CPC; van den Oord, Li, and Vinyals, 2018) trains a context representation to identify a future observation among sampled alternatives. Under its stipulated sampling scheme, InfoNCE supplies a lower bound on predictive mutual information; it does not impose the explicit input-rate penalty of IB. Appendix C states the objective, density-ratio interpretation, and sampling assumptions. These frameworks therefore supply comparisons whose relevance to the present hypothesis can be assessed without assuming that hippocampal learning instantiates either algorithm.

The key geometrical limitation can be stated directly. CPC’s log-bilinear score is $f_k(u, c) = \exp\{Z(u)^\top W_k c\}$. For an invertible matrix B , replacing $Z(u)$ by $BZ(u)$ and W_k by $B^{-\top}W_k$ leaves every predictive score unchanged while generally changing Euclidean and cosine distances among targets. Thus functional equivalence does not fix representational geometry, and improving InfoNCE does not by itself predict the published negative local distance correlation. The same qualification applies to claims that increased predictive information entails increased neural distance.

The experimental question is narrower: whether making particular alternative interpretations available changes held-out discrimination or neural differentiation at fixed target and independently normed consequence. Recently activated interpretations, alternative referents, and recalled episodes are possible sources of competition. A positive competitor-availability effect would establish functional sensitivity without identifying biological negative sampling. The behavioral test of consequential distinction remains intelligible if CPC is rejected as a model of the underlying neural process.

8 Experiments that distinguish the interpretations

A controlled follow-up should vary contextual distinction independently of lexical similarity. Closely related expressions could appear in contexts where substituting one for the other either preserves or changes the interpretation required by a subsequent comprehension question. Frequency, duration, phonetic similarity, and syntactic structure would need balancing or explicit modeling. The central outcome would be whether neural separation tracks the contextual consequence of substitution beyond what the comparison embedding predicts. This design would translate the integration–separation question exemplified by Schlichting et al. (2015) into a language paradigm, while testing a different manipulation. Where anatomy and sample size permit, separate analyses along the hippocampal long axis would help avoid averaging across potentially distinct coding regimes.

The primary operationalization of the probe-relative substitution consequence would use an independent behavioral sample, collected and fixed before the neural data are analyzed. For each item pair and context condition, norming participants would receive either the intended interpretation or an explicitly specified substituted interpretation, followed by the same balanced set of forced-choice comprehension probes. The answer key would be defined from the narrative content. Define $q_{ij}^{(k)}$ as the proportion of probes for which the normed answer under the substituted interpretation differs from the answer under the intended interpretation, averaging both substitution directions where appropriate. Agreement thresholds and exclusion rules would be prespecified, with poorly understood or inconsistently interpreted items revised or removed before neural recording. This measure captures a task-relative consequence of substitution, not subjective severity or a universal semantic cost. It involves no neural measurement and does not use embedding distance to define consequentiality.

The item-pair-to-occurrence assignment used to construct Q must be fixed before neural analysis. Norming uncertainty should be retained through resampling of norming participants and probes, or through an explicitly specified measurement-error model. Probe selection is part of the estimand: a consequence score depends on which comprehension questions are sampled. Balancing probe types and testing robustness to alternative prespecified probe sets would limit dependence on an idiosyncratic answer key.

Writing the consequence score explicitly clarifies its domain. Let $A_{i \rightarrow j, l}^{(k)}$ and $A_{i, l}^{(k)}$ be the independently normed categorical answers for probe l under a substituted and intended interpretation, respectively. For L prespecified probes, define

$$q_{i \rightarrow j}^{(k)} = \frac{1}{L} \sum_{l=1}^L \mathbf{1}\{A_{i \rightarrow j, l}^{(k)} \neq A_{i, l}^{(k)}\}, \quad Q_{ij}^{(k)} = \frac{q_{i \rightarrow j}^{(k)} + q_{j \rightarrow i}^{(k)}}{2}. \quad (11)$$

The symmetric definition requires both directions to be meaningful and normed under the specified comparison design; otherwise the pair should be excluded from this analysis or a prespecified alternative estimand used. A score of one means every sampled answer changes, not that confusion is maximally harmful in every possible context. Normed categorical answers should be accompanied by agreement estimates, since a modal answer can conceal ambiguity. Probability-based answer-distribution contrasts could provide a separate sensitivity measure, but their choice and normalization must be declared in advance. Consequence, confusion probability, and subjective severity are distinct quantities.

A distinct norming block would present the original speech without instructing a substitution and measure the rate of foil-consistent comprehension errors, with reaction time recorded as a secondary index of difficulty. These outcomes estimate susceptibility and processing demand separately from q .

In the recording sample, the same probe types would verify comprehension and supply a behavioral outcome; they would not redefine q after the neural results are known. The primary analysis would test whether the prespecified consequence measure predicts held-out neural differentiation after accounting for independently estimated confusion susceptibility and difficulty. Matching attention and task demand remains essential, because a consequence effect alone would not distinguish semantic differentiation from more general task engagement.

A fuller test of the compression–relevance interpretation would manipulate relevance separately from the availability of a distinction. The same recordings could be presented under counterbalanced task instructions that make a particular semantic contrast informative for the forthcoming probe in one condition and irrelevant in another. Independent norming should establish both that the interpretations remain discriminable under either instruction and that their contribution to the declared outcome changes. Merely selecting different probes after seeing neural responses would not constitute this manipulation. Instructions, probe distributions, incentives, and analysis contrasts must be fixed in advance, with attention and difficulty assessed independently. Event-script framing provides a precedent for altering relevance while preserving the stimulus, but does not remove these alternative explanations.

For a prespecified matched pair, let $D_{s,ij}(g)$ denote neural distance under randomized relevance assignment $g \in \{0, 1\}$. A participant- and stimulus-aware analysis could test the assignment contrast

$$\Delta_{\text{rel}} = \mathbb{E}_{s,(i,j)}[D_{s,ij}(1) - D_{s,ij}(0)]. \quad (12)$$

A positive contrast accompanied by improved held-out recovery of the task-relevant interpretation would support relevance-sensitive differentiation under the declared metric and readout. It would not demonstrate IB optimization: increased distance is only one possible implementation of improved recovery, and a complete bottleneck test would also require an identified measure of compression and explicit model comparison. The narrower consequence experiment tests whether independently normed substitution consequences explain usable differentiation; the relevance extension asks whether changing the task’s predictive target changes which distinctions are retained. These are related hypotheses with distinct estimands, and their randomization tests must follow their actual assignment units rather than permuting pairwise cells.

The distribution-preservation criterion of Section 4.1 can be evaluated in a controlled repeated-condition extension. Estimate the independently normed outcome distribution for each eligible condition under the same fixed probe protocol, including probe identity in the outcome when answers to different questions cannot be pooled meaningfully. Fit a probabilistic neural readout within training folds, then average its predictions over held-out responses to estimate $\hat{\rho}_i$. Endpoint-wise total-variation errors and their participant-, stimulus-, and norming-aware uncertainty can be compared with a prespecified tolerance. The criterion provides an interpretable task-distinction bound only where both endpoint errors are sufficiently small. Trials supporting response averaging must be repetitions of a declared stimulus condition or draws from a justified condition ensemble, not silently pooled distinct narrative occurrences. Separately evaluate trial-wise interpretation or answer prediction against a baseline that has no neural response input. Classification accuracy alone does not estimate total-variation calibration, and this readout extension does not redefine the substitution score Q from neural data.

This first experiment tests contextual discriminability. Relational binding would require a separate manipulation, such as assigning matched occurrences to the same or different narrative episodes while holding substitution consequences constant, together with a probe of memory for that episode association. Such a factorial extension could ask whether binding modulates discrimination,

but it is not necessary to the narrower claim and is not inferred from distance measurements alone.

A complementary experiment could hold an ambiguous word constant while varying cues that establish different interpretations. Such a design would separate lexical identity from contextual differentiation and allow repeated presentations within each condition. Independent behavioral judgments would help distinguish successful disambiguation from neural variability without a demonstrated semantic consequence. Recording across the relevant sequence could also clarify whether separation follows the disambiguating cue, precedes it through expectation, or emerges through later integration.

Acoustic and lexical baselines should be specified at the occurrence level before pairwise contrasts are constructed. Candidate controls include intensity, pitch and prosodic prominence, phoneme identity and timing, coarticulation, lexical frequency, cohort size, phonological neighborhood density, and contextual surprisal, alongside duration and syntactic structure. Frequency is a lexical distributional measure, whereas surprisal is conditional on a specified predictive model and context; they cannot be substituted for one another. Closely adjacent response windows may also share acoustic or neural activity, requiring temporal-separation controls and declared overlap exclusions. Where feasible, identical target recordings embedded in counterbalanced contexts would reduce target-acoustic variation, while preserving natural prosody would require explicit checks rather than assuming splicing is innocuous.

These controls do not make semantics statistically pure. Semantic, syntactic, and predictive features covary, and indiscriminately orthogonalizing embeddings against acoustic or surprisal predictors can remove task-relevant structure or make conclusions depend on residualization order. Baseline and consequence-augmented models should instead be compared on the same held-out data; any residualization must be fitted within training folds, with its direction declared. Unique predictive improvement is conditional on the included baselines, not an exhaustive attribution of residual variance to semantic distinction. Because surprisal and task engagement may mediate effects of context, analyses with and without them should distinguish a conditional association from the total effect of a randomized context manipulation.

Analytical robustness is equally necessary. Similarity ranges and distance measures should be prespecified using an independent stimulus set or selected in a training partition and evaluated in held-out narratives. Report the full association alongside local slopes and their uncertainty, including sensitivity to shifted range endpoints and a continuous nonlinear model that does not require a hard boundary. Repetition across BERT and GPT-2 is informative, but ranges should be calibrated separately for each representation rather than assuming that equal numerical distances imply equal semantic similarity. A sign change that depends narrowly on one endpoint choice would weaken the interpretation of a distinct coding regime.

Pairwise comparisons reuse occurrences and cannot be treated as millions of independent biological replications. One concrete approach is to estimate participant-specific slopes or contrasts and summarize them with participant-level uncertainty. Another is a multilevel model with participant effects and crossed occurrence effects for both members of a pair, supplemented by stimulus or narrative effects where the design permits. Label permutations must act on the occurrence-level assignment and regenerate the full distance matrix, preserving shared endpoints, rather than independently shuffling pairwise cells. For naturalistic speech, the permutation scheme should respect exchangeability through matched blocks or a justified temporal procedure; arbitrary shuffling can destroy narrative dependencies. Hierarchical bootstrap resampling of participants and narrative blocks provides a complementary sensitivity analysis. These recommendations specify alternatives for future testing and do not imply that the published analyses omitted all such controls.

The permutation null must match the regression claim. Shuffling occurrence labels across a whole narrative tests a broad alignment null and generally disrupts the legitimate relations among Q , embeddings, and nuisance features; it is not automatically a valid test of the adjusted Q coefficient. In a randomized experiment, the primary randomization test would reassign the context or consequence manipulation at its actual assignment unit within the permitted blocks. For every reassignment, one must reconstruct affected occurrence labels and pair-level predictors and refit the same preregistered analysis, including any data-dependent steps permitted within training folds. A whole-matrix label permutation, when exchangeable, must use the same permutation for rows and columns rather than permuting cells independently. If several participants receive the same randomized item assignment, the test must preserve that shared assignment structure rather than independently permuting each participant.

Where Q varies observationally and no defensible assignment or exchangeability scheme exists, an exact permutation test of its conditional effect should not be claimed. Participant- and stimulus-aware uncertainty estimation, a calibrated model-based bootstrap, and held-out prediction are alternatives with their own assumptions. Stimulus blocks shared across participants should be resampled jointly in a crossed participant–stimulus procedure, preserving endpoint identities and reconstructing distances, rather than resampled independently within each participant. With few narratives, stimulus-generalization uncertainty may remain poorly determined. Null simulations should assess type-I error under shared-endpoint noise and temporal dependence, and the primary coefficient or contrast, local-range rule, multiplicity adjustment, and smallest effect of interest should be fixed before testing. An inconclusive interval is not evidence against the hypothesis; a sufficiently precise interval excluding the prespecified meaningful positive effect would provide the relevant negative result.

A reproduction across independently specified ranges and measures would strengthen the geometrical claim, while a relation between experimentally manipulated confusion consequences and held-out neural differentiation would strengthen the proposed functional extension. Learning-related change or intervention would be needed to identify the mechanism. This separates three possible achievements: reproducing a sign change, explaining its relation to comprehension, and establishing the process that produces it.

9 Conclusion

The hippocampal study offers evidence that semantic organization during narrative listening is distributed, contextual, and not uniformly captured by a simple correspondence between embedding distance and neural distance. Its theoretical interest lies in the conjunction of broad relational structure with local differentiation. A listener must preserve the affinities through which expressions become intelligible and the distinctions through which their present interpretation remains determinate.

The contextual-discrimination hypothesis proposed here treats those requirements as complementary constraints on representation, with relational memory providing motivation rather than an inferred binding mechanism. Context changes which similarities can support generalization and which differences must remain recoverable. The reported local sign change makes this proposal worth testing, while leaving its adaptive and mechanistic explanations open. A mature account of semantic geometry would therefore connect population organization to independently measured discrimination and comprehension, explaining how continuity across contexts can coexist with the preservation of consequential differences.

A Predictive distortion and the information bottleneck

The information-bottleneck framework permits an exact statement of why similar inputs may need distinct predictive codes. Let $p(x, y)$ be a joint distribution, and let a stochastic encoder $p(t | x)$ satisfy the Markov condition $Y - X - T$, meaning that the representation is generated from X without additional access to Y . For discrete variables define

$$d_{\text{IB}}(x, t) = D_{\text{KL}}(p(Y|x) \| p(Y|t)), \quad D_{\text{KL}}(p \| q) = \sum_y p(y) \log \frac{p(y)}{q(y)}. \quad (13)$$

The divergence is directed and can be infinite when q assigns zero probability to an outcome having positive probability under p . It is not a metric between neural vectors. Its expectation satisfies

$$\mathbb{E}_{X,T} d_{\text{IB}}(X, T) = H(Y|T) - H(Y|X) = I(X; Y) - I(T; Y) \geq 0. \quad (14)$$

Expanding the expected logarithm and using conditional independence gives the first equality; the definition of mutual information gives the second. The quantity measures lost predictive information, equivalently excess expected logarithmic prediction loss, under the stated model. It cannot be equated with cosine distance or applied to arbitrary neural observations without a probabilistic specification.

Tishby et al.'s stationary encoder equation makes the role of distortion explicit:

$$p(t|x) = \frac{p(t)}{Z_\beta(x)} e^{-\beta d_{\text{IB}}(x,t)}, \quad Z_\beta(x) = \sum_t p(t) e^{-\beta d_{\text{IB}}(x,t)}. \quad (15)$$

The representative prediction $p(y | t)$ depends on the encoder and joint distribution, so this is a self-consistency condition. At fixed representative predictions, assigning an input to a code with greater predictive distortion receives less probability, other factors equal. Divergent predictions can therefore motivate retaining different codes, but they do not require those codes to have greater neural Euclidean or cosine distance.

For an elementary example, let X take equally probable values a and b , with $p(Y = 1 | a) = \epsilon$ and $p(Y = 1 | b) = 1 - \epsilon$ for $0 < \epsilon < 1/2$. Merging them into a single code gives $p(Y = 1 | t) = 1/2$ and loses predictive information

$$\log 2 - H_b(\epsilon), \quad H_b(\epsilon) = -\epsilon \log \epsilon - (1 - \epsilon) \log(1 - \epsilon). \quad (16)$$

A deterministic two-code representation removes that loss while raising $I(X; T)$ from zero to $\log 2$. Comparing only these two candidate encoders, retention is favored when β times the lost predictive information exceeds $\log 2$. Stochastic encoders can produce intermediate trade-offs, so the comparison is not a general phase-transition result. It illustrates why arbitrarily close comparison embeddings can conceal consequential predictive differences, without deriving any neural-distance sign change.

Tishby et al. also discuss bifurcations of code solutions as β changes in the information plane. Such changes concern predictive quantizations and code cardinality. Franch et al.'s reduced-rank regression instead estimates linear predictive dimensions for different embedding models. Rank 12 for GPT-2 and rank 3 for Word2Vec therefore do not demonstrate bottleneck bifurcations or

hierarchical code splitting. Connecting the quantities would require a fitted generative model relating code structure to regression rank.

Futrell and Hahn (2026) offer a related but distinct sequential-processing account. They define predictive information as dependence between a process’s past and future and show that codes constrained to have low predictive information can express approximately independent meaning features systematically and locally, with cross-linguistic corpus comparisons supporting their structural argument. This concerns the organization of linguistic signals under processing constraints. It is not evidence that a hippocampal representation minimizes the IB objective, and minimizing predictive dependence within a signal is not equivalent to maximizing CPC’s predictive-information bound. The contribution is a precedent for deriving structural hypotheses from explicitly specified information constraints rather than inferring an objective from alignment alone.

B Adjustment and predictive improvement

In an ordinary linear regression with independent errors, the conditional coefficient can be written using projections. For N eligible observations, let $B \in \mathbb{R}^{N \times p}$ be the baseline design matrix and $y, q \in \mathbb{R}^N$ the outcome and consequence vectors. Write $P_B = BB^+$ for the orthogonal projection onto the column space of B , where B^+ is its Moore–Penrose pseudoinverse, and set $M_B = I_N - P_B$. Both projection operators are $N \times N$; when B has full column rank, $P_B = B(B^\top B)^{-1}B^\top$. Provided $q^\top M_B q$ is positive,

$$\hat{\beta}_Q = \frac{q^\top M_B y}{q^\top M_B q}. \quad (17)$$

This expression shows why nuisance adjustment applies to both the outcome and the consequence predictor: it compares their components not accounted for by B . Residualizing y alone and regressing it on unadjusted q is generally a different procedure. The expression is explanatory; the proposed hierarchical analysis requires covariance-aware estimation and cannot import independent-error standard errors from this elementary case. Small residual variation in q makes estimation unstable even when its raw variation is substantial.

Predictive improvement can be stated independently of coefficient significance. For held-out eligible pairs P_{test} , define

$$\Delta_{\text{pred}} = \frac{1}{|P_{\text{test}}|} \sum_{p \in P_{\text{test}}} \left[(y_p - \hat{y}_{0p})^2 - (y_p - \hat{y}_{1p})^2 \right], \quad (18)$$

where model 0 contains the declared baselines and model 1 additionally includes consequence and the prespecified interaction. Positive improvement indicates lower test error for the augmented model. Pairwise test losses remain dependent, so uncertainty must respect their participant and stimulus structure. Weighting should also be specified: equal weighting of pairs favors participants or narratives contributing many eligible comparisons, whereas averaging participant-specific losses targets a different population summary. Coefficient direction, predictive gain, and behavioral discrimination are complementary criteria rather than interchangeable evidence.

C Contrastive predictive coding: objective and assumptions

Contrastive predictive coding (CPC; van den Oord, Li, and Vinyals, 2018) offers a predictive-learning comparison whose assumptions can be stated explicitly. Its encoder constructs latent observations, while an autoregressive network summarizes their history into a context representation. To avoid confusing the CPC notation with the commentary’s situated context, write this learned summary as C_t and a future observation as $U = x_{t+k}$. For each context, a candidate set contains one positive observation drawn from $p(u | C_t)$ and $N - 1$ independent alternatives drawn from the marginal $p(u)$. With a positive scoring function f_k , its population objective and information bound are

$$\mathcal{L}_{N,k} = -\mathbb{E} \log \frac{f_k(U^+, C_t)}{\sum_{j=1}^N f_k(U_j, C_t)}, \quad I(U; C_t) \geq \log N - \mathcal{L}_{N,k}. \quad (19)$$

The expectation is over the joint context–positive distribution and the stipulated negative-sampling distribution. The bound concerns that population objective, rather than every finite-sample estimate, and its maximum possible value is $\log N$. It is a lower bound on predictive information, not an estimate of the information rate compressed from the entire speech history. In particular, CPC does not include the explicit input-rate penalty in the IB objective. Architectural restrictions may constrain capacity, but a compact vector and a penalized mutual-information rate are different commitments. Thus IB and CPC supply related perspectives on predictive retention without constituting equivalent optimization problems.

The density-ratio interpretation follows from the candidate-identification problem. When the positive index is uniformly distributed, the Bayes posterior is proportional to $p(U_j | C_t)/p(U_j)$. An unrestricted optimal score consequently recovers this ratio up to a context-dependent multiplicative factor. CPC uses a log-bilinear critic; writing its target latent vector as $Z(u)$ gives

$$f_k(u, c) \propto \frac{p(u | c)}{p(u)}, \quad f_k(u, c) = \exp\{Z(u)^\top W_k c\}. \quad (20)$$

The first relation describes the ideal density-ratio solution, whereas the second specifies a restricted parameterization that need not attain it. Neither relation is a metric between two word-response vectors. Indeed, replacing $Z(u)$ by $BZ(u)$ for an invertible matrix B and W_k by $B^{-\top}W_k$ leaves every score unchanged while potentially altering Euclidean and cosine distances among target vectors. This invariance parallels the mapping concern in Section 5: equivalent predictive scores do not determine a unique distance geometry. A competitor can receive a lower score through a changed critic or context representation without requiring increased pairwise neural distance. Under these assumptions, optimizing InfoNCE therefore does not predict a negative local semantic–neural distance correlation.

The selection of alternatives is equally consequential. The standard bound assumes marginal negatives. Sampling only semantically similar alternatives, or excluding same-episode observations, changes the candidate distribution and requires a correspondingly derived objective; the original mutual-information interpretation cannot simply be carried over. False negatives are also possible when two observations preserve the same relevant interpretation. CPC predicts sampled future observations, whereas the commentary’s behavioral variable concerns consequences for declared comprehension probes. Future predictability can retain acoustic regularities or other features without preserving the intended task distinction, so identifying these targets would require evidence rather than terminology. BERT and GPT-2 in the hippocampal comparison are not thereby shown to have been trained with InfoNCE.

The biological question is what would select the alternatives. Candidate competitors include recently activated interpretations, alternative referents, or recalled episodes sharing cues. Experimentally manipulating which alternatives are available, while holding the target and its independently normed consequence score constant, could test whether differentiation depends on competitor availability. Such an effect would support a functional role for competition without identifying negative sampling, a density-ratio critic, or InfoNCE optimization in neural circuitry. The joint information-theoretic interpretation is therefore deliberately conditional: IB specifies which declared predictive distinctions compression should preserve, and CPC illustrates one way to train predictive scores against sampled alternatives. A model that connects either objective to the reported hippocampal geometry must additionally specify its representation, noise, readout, and competitor distribution.

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